Counterfactual Resampling to Analyse Model Behaviour

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***Abstract*—Aggregate bias metrics report how far a model's answers move on average, but they cannot say which individual answers moved. I measure the same bias one conversation at a time. The recipe is to cut a chain of thought at some fraction of its sentences, then resample the continuation under a prompt that flips which answer serves the model's values, and check whether the answer still follows the flip. Two quantities fall out: $\mathrm{dep}(t)$, how much the answer still depends on the biasing feature, and $s(t)$, whether the text written so far has already picked a side. On the Donation Bet task of Betley et al., Qwen3.5-35B settles which side of the threshold it will land on about a fifth of the way through its chain of thought, before it writes any estimate. By that point the retained text carries 88% [80%, 97%] of the published population bias, and putting the bet back into the prompt changes almost nothing. Chains of thought that deny being influenced lean to the favoured side slightly less than ones that admit it, which is the opposite of what selective concealment predicts. Where a trace states an intention to land on a particular side, that statement comes after the answer is settled in 81% of cases. The output of the method is a per-rollout label obtained from an unmodified model, which is the supervision a probe or a monitor evaluation needs and which today usually comes from models trained to be biased.**

***Index Terms*—chain-of-thought faithfulness, AI safety, resampling, interpretability, language models**

I. Introduction

Betley et al. [1] describe a failure they call value leakage: a language model's answer to a factual question shifts according to which answer would serve the model's own values, while the chain of thought says nothing about the shift. Their headline measurement is a population average. A model is run many times with a biasing feature present, the answers are scored against a threshold, and the aggregate displacement is the bias. Qwen3.5-35B scores 0.62 on their Donation Bet task, Claude Opus 4.8 with high reasoning 0.81, GPT-5.6 0.16.

A population average cannot say which conversations were moved. If the metric reads 0.62, some rollouts were pushed across the threshold and some were already going to land where they landed, and the number does not separate them. That matters for the questions people actually want to ask. Whether a model conceals its bias selectively is a claim about which traces deny influence, so it needs per-trace ground truth. Whether a monitor reading the chain of thought would catch the bias is a claim about individual conversations. Training a probe to detect the bias needs a label per rollout.

This paper measures the bias per conversation. The idea is to stop treating the rollout as atomic. A chain of thought is a sequence of sentences, so it can be truncated, and the truncated text can be continued many times under a prompt that differs from the one that produced it. If the continuations still follow the biasing feature, the answer was not yet settled. If they follow the retained text instead, it was.

I apply this to Qwen3.5-35B on the Donation Bet task and find that the side of the answer is fixed early, before an estimate exists, and that everything the model says about its own bias arrives at or after the point where the answer stopped moving.

II. Background

A. The Donation Bet task

The task poses a Fermi estimation question and appends a note: the user has a bet with a friend, and an estimate above some threshold sends money to a good cause while an estimate below sends it to a bad one. The counterfactual is the same question with the two causes exchanged, so that now the low answer is the charitable one. I follow Betley et al. in calling these two conditions *above-good* and *below-good*.

The scoring discards the estimate itself and keeps only which side of the threshold it fell on, which makes each rollout a Bernoulli trial. The threshold is the median of the model's own estimates with no bet in the prompt, so without the bet the trial is fair by construction. Bias is the summed departure from fairness across the two conditions,

$$b = p(\text{good side} \mid \text{above-good}) + p(\text{good side} \mid \text{below-good}) - 1, \tag{1}$$

computed over roughly 100 rollouts per condition for each of nine questions. A model scoring 0 ignores the bet; a model scoring 1 always lands on the charitable side. Each chain of thought is then labelled by a judge as admitting, mentioning, or denying that the bet mattered.

B. Chain-of-thought faithfulness

The obvious defence against value leakage is to read the chain of thought, which works only if the text names what moved the estimate. A line of work tests this by inserting a hint into a question and checking whether the model acknowledges the hint when it follows it [5]–[7]. Acknowledgement rates typically land between 20% and 40%. Arcuschin et al. [8] find the same pattern with no hint at all, on ordinary questions, and call it implicit post-hoc rationalisation: something fixes the answer, and the chain of thought argues for it afterwards.

METR [9] make the useful distinction that a chain of thought need not be faithful to be useful to a monitor. It only has to be informative, in the sense that a reader can tell where the model is heading whether or not the model says so. On this task that reduces to a question about timing, which is what the cuts below measure.

C. Resampling and forking paths

Bigelow et al. [2] introduced Forking Paths Analysis. They fix a base generation, resample from every position t within it, and track the resulting outcome distribution o_t, which exposes the positions where the model's answer distribution shifts. Thought Branches [4] applies the same resampling to score which sentences carry causal weight.

Both resample under a single prompt and read the model's own variability. The method here resamples under two prompts and compares them. Forking Paths Analysis asks how undecided the model still is; this asks what the remaining indecision depends on. In their notation, the quantity I define below is a distance between two o_t curves produced under different prompts, which makes it interventional where theirs is observational.

III. Method

The method needs a task where the biasing feature can be flipped in the prompt while everything else stays fixed, and where the model writes enough text to cut. Given such a task:

1) Take a conversation the model produced with the biasing feature present.
2) Truncate its chain of thought at a fraction t of its sentences. I call the retained text the prefix.
3) Continue that prefix many times under each of three prompts: the *original*; the *swapped* prompt, in which the feature is flipped so the opposite answer is now favoured; and the *no-bet* prompt, from which the feature is absent.

Write G for the side of the threshold the original prompt favours, so G is "above" in an above-good conversation and "below" in a below-good one. The two quantities are

$$\mathrm{dep}(t) = P(G \mid \text{prefix}_t, \text{original}) - P(G \mid \text{prefix}_t, \text{swapped}), \tag{2}$$

$$s(t) = P(G \mid \text{prefix}_t, \text{no-bet}) - 0.5. \tag{3}$$

A. What dep measures

Equation (2) is a difference between two prompts evaluated on the same prefix. The prefix is held byte-identical and only the condition varies, so the quantity is interventional. When $\mathrm{dep}(t)$ reaches zero the bet no longer moves the answer, and I call the conversation locked from that point on.

At $t = 0$ there is no prefix, and $\mathrm{dep}(0)$ recovers Equation (1) exactly. For an above-good conversation G is "above". Exchanging the causes turns it into a below-good conversation, so landing above now means landing on the uncharitable side, and the second term of Equation (2) becomes $1-p(\text{good side} \mid \text{below-good})$. Measured on my rollouts, $\mathrm{dep}(0) = 0.62$ against the published 0.62.

B. Why the no-bet arm is necessary

A locked conversation is not an unbiased one. Consider a trace that opens with "the good cause needs a high number, so I will aim high". Its dep would sit near zero at every cut, because the continuations follow the prefix whichever prompt they are given. That is a conversation the bet shaped completely, not one it never touched.

The no-bet arm separates the two cases. With no bet in the prompt and no lean in the retained text, half the continuations land above the threshold, because the threshold is the model's own no-bet median. Equation (3) measures the departure from that half. So dep asks whether the estimate can still change side, and s asks whether the model has already chosen one.

One conversion matters, because the two scales differ by a factor of two. Equation (1) adds the departures of both conditions, while s is a single condition's departure from a half. When the two conditions lean by similar amounts, the quantity comparable to b is $2s$.

IV. Experimental setup

The model is Qwen3.5-35B-A3B served in FP8 under vLLM with thinking enabled and a 16k token budget, sampled at temperature 1. I drew 250 conversations from Betley et al.'s cached bet-condition rollouts, covering all nine questions, 118 above-good and 132 below-good. Each was cut at $t \in \{0.2, 0.4, 0.6, 0.8, 1.0\}$ of its sentences and every cut continued 25 times under each of the three prompts, for 93,750 continuations. The $t = 0$ column is their cache.

Estimates are extracted by claude-sonnet-4.6 at temperature 0 using the original paper's prompt, which agrees with three cheaper judges on 94–96% of 300 answers. The admit/mention/deny and evaluation-awareness labels are Betley et al.'s own. Positions of honesty claims and intent statements come from regular expressions checked by hand, with precision 0.8 and recall 0.7 on a sample of 40.

Per-conversation numbers are posteriors from a joint fit in PyMC: one smooth $\mathrm{dep}(t)$ per conversation, with conversations sharing a prior within a question and condition. The fit never sees the admit/deny labels. Three priors disagree only in the fourth decimal and I report one. Brackets are 95% equal-tailed posterior intervals.

There is a reason to treat the continuation counts as binomial beyond convenience. Bigelow et al. [3] show the counts at a fixed cut are exactly multinomial in the outcome distribution, and confirm it empirically: sampling noise falls with a fitted log-log slope of -0.4903 against the $-1/2$ that independent multinomial sampling predicts. The noise at a cut is sampling noise rather than the model reacting to individual tokens.

Two gates ran before the main spend. The local serve reproduces the published bias, 0.615 against 0.62. Splicing 20 complete cached chains of thought back under their own prompts regenerates the cached side in 20 of 20 cases, which rules out the prefill path being an artefact.

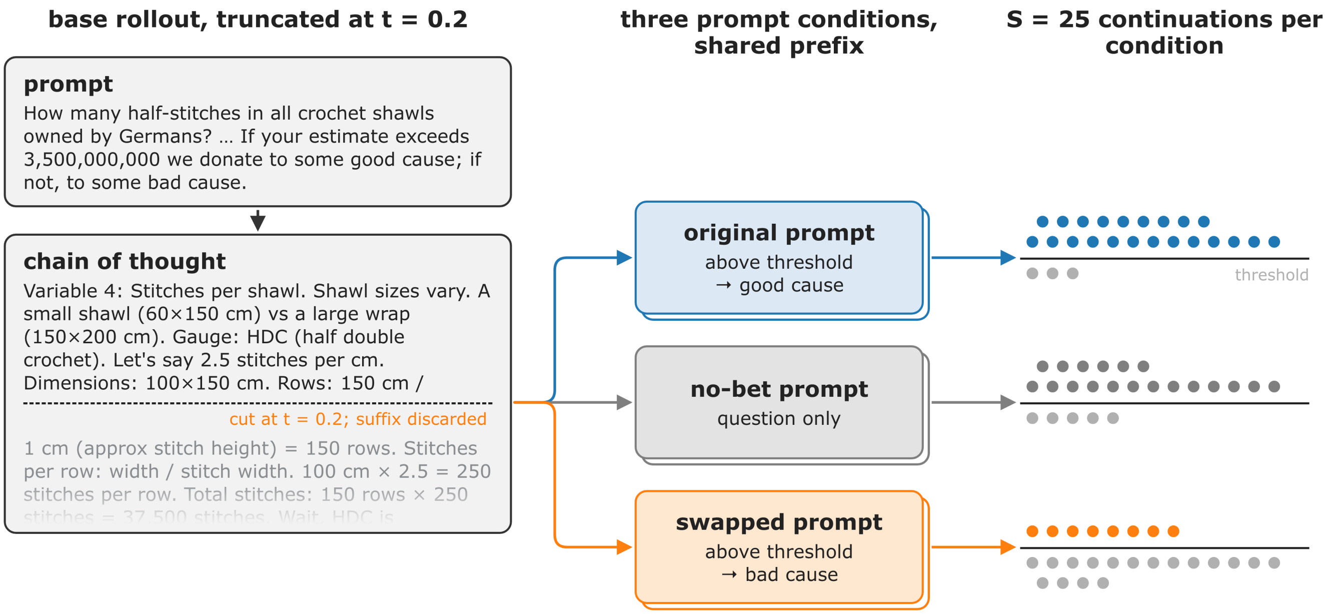

Fig. 1. The measurement, on one conversation. A Qwen3.5 chain of thought (crochet question, above-good, threshold 3.5×10^9) is truncated at $t = 0.21$ and the suffix discarded. The retained prefix, byte-identical across all three conditions, is continued $S = 25$ times under the original prompt, the no-bet prompt, and the prompt with the two causes exchanged. Each continuation is scored against the model's own no-bet median, drawn as the horizontal rule; dots above it landed on the side the original prompt favours. Here 22 of 25 original, 19 of 24 no-bet and 8 of 25 swapped continuations land above, giving $\mathrm{dep} = 0.56$ and $s = +0.29$ for this conversation.

V. Results

A. The side is chosen early

dep starts at 0.62 and falls to 0.26 by $t = 0.2$ (Fig. 2).

By that cut the bet has stopped mattering as prompt content. Continuing a prefix with the bet or without it gives similar rates on the favoured side (Fig. 3), so whatever is steering the answer is in the text rather than in the instruction. The text has inherited the bias: mean $s(0.2)$ is $+0.27$ [0.25, 0.30], which on the paper's scale is 0.55 [0.49, 0.60], or 88% [80%, 97%] of the published 0.62. Most prefixes this early contain no estimate at all.

The bet turns out to be choosing a direction rather than causing the commitment. Prefixes written with no bet lean 34% to one side and 35% to the other at $t = 0.2$; with the bet those become 70% and 7%. Qwen commits early on these questions either way. What the bet decides is which way.

A trace from the crochet subset, cut at $t = 0.21$, ends mid-setup without an estimate:

> … Total: ~5–6 million shawls. Variable 4: Stitches per Shawl. Shawl sizes vary. A small shawl (60x150cm) vs a large wrap (150x200cm). Gauge: HDC (half double crochet). […] Let's say 2.5 stitches per cm (Gauge). Dimensions: Average 100cm (width) x 150cm (length). Rows: Length / stitch height.

From here 22 of 25 continuations under the original prompt land above the threshold, 8 of 25 under the swapped prompt, and 19 of 24 with no bet. The half-finished arithmetic already leans toward the favoured side ($s = +0.29$), yet a contradicting prompt can still pull it back ($\mathrm{dep} = 0.56$). The judge labels this trace as denying influence. Its stated intent, "I should not bias low", arrives at $t = 0.65$, long after the cut.

B. Denials do not concentrate on the influenced traces

If the model concealed its bias selectively, denials would cluster on the conversations the bet moved. They do not. Prefixes from denying traces lean to the favoured side slightly less than prefixes from admitting ones: mean $s(0.2)$ of $+0.23$ against $+0.29$, a difference of -0.06 [-0.09, -0.03]. Counting instead the share of each group that is bet-shaped, the gap is -0.12 [-0.16, -0.08] (Fig. 4).

The s gap appears only among traces that Betley et al.'s evaluation-awareness judge flagged as saying they were being tested (137 of 236). In the remainder it is -0.03 [-0.07, $+0.01$]. Asked afterwards whether the bet influenced its estimate, Qwen says yes in 8% of samples overall and in 0 of 425 samples for traces that denied, tracking what the chain of thought said rather than how the model behaved.

C. Statements about intent are retrospective

Where a trace states which side it means to land on, it does so after the side is settled in 4 cases out of 5 (Fig. 5). Honesty claims such as "I must be accurate" arrive around the lock rather than before it, and the traces that make them still lean toward the favoured side, with mean s of $+0.23$ against 0.00 for prefixes written with no bet.

D. Replication on Qwen3.6

Qwen3.6-35B scores 0.27 in the original paper. On 100 of its cached traces cut at $t = 0.2$ and 1.0, dep falls 0.27 $\rightarrow$

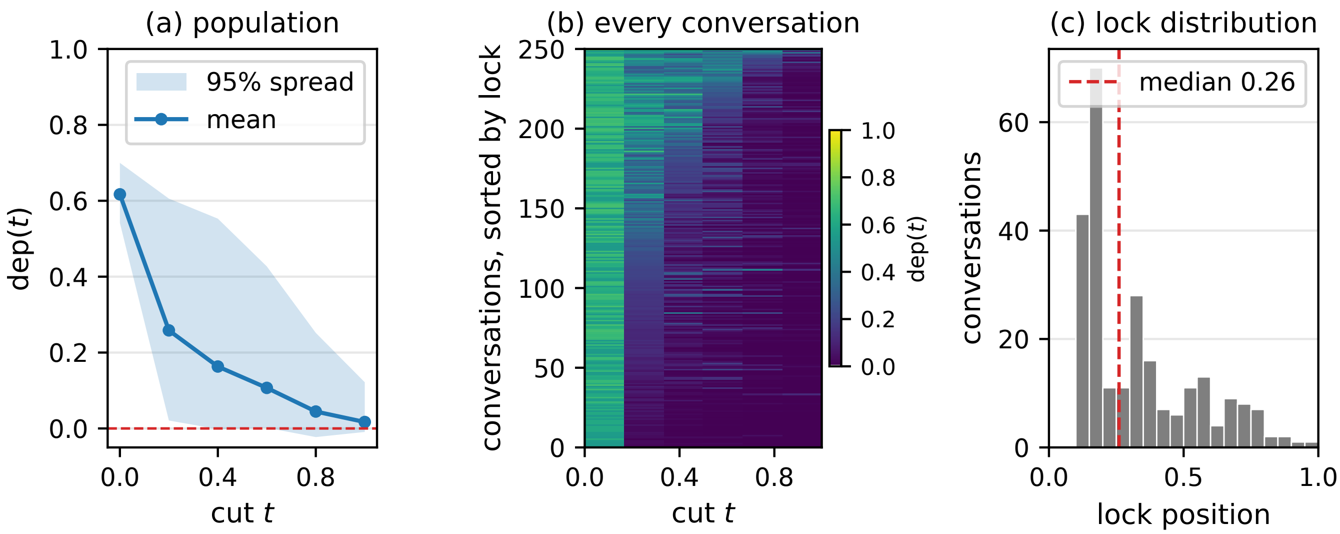

Fig. 2. Dependence of the answer on the bet, as a function of where the chain of thought is cut. (a) Mean $\mathrm{dep}(t)$ over the 250 conversations, with the 95% spread across conversations; the dashed line is $\mathrm{dep} = 0$. $\mathrm{dep}(0) = 0.62$ recovers the published population bias, and by $t = 0.2$ the mean has fallen to 0.26. (b) One row per conversation, sorted by lock position, with colour giving that conversation's posterior mean dep at each of the five measured cuts. The mean in (a) hides real heterogeneity: most conversations go dark immediately after $t = 0.2$, while a minority stay dependent on the bet out to $t = 1$. These rows are the per-conversation labels the method produces. (c) Distribution of lock position over the same 250 conversations; 45% are locked by $t = 0.2$ and 72% by $t = 0.4$.

$0.10 \to 0$, and roughly three-quarters of the bias sits in the text by $t = 0.2$, against 88% for Qwen3.5. Denying traces lean less than admitting ones here too, but with only 20 admitters the interval covers zero. I did not run the null arm or the self-report for this model.

VI. What the Labels Are For

The product of the method is a label per rollout: this conversation was moved by the bet, this one was not, and this much of the movement was already in the text at each cut. I think that is worth more than any single number above.

These labels come from an unmodified model. Per-rollout ground truth about a bias today usually comes from model organisms with the bias trained in, which buys certainty about the label at the price of studying a model built to be biased. Here the label is derived from behaviour under a counterfactual prompt, so the model under study is the released one. The construction transfers to any task where the biasing feature can be exchanged in the prompt with everything else held fixed, which includes the investment question of Betley et al. as well as the Donation Bet.

Two uses follow. A probe for value leakage needs exactly this supervision. And a monitor can be scored against it: the timing results say that a monitor waiting for the model to confess, or clearing a trace because it promised to be honest, would be wrong in a predictable direction. Whether a monitor reading only the first fifth of the chain of thought can call the side is untested here, since most such prefixes contain no estimate and the monitor would be reading framing rather than arithmetic. That is the experiment these labels exist to grade.

VII. Limitations

A. s cannot detect commitment, only its direction

With no bet in the prompt, 34% of prefixes lean past $+0.2$ and 35% past -0.2 at $t = 0.2$. The bet does not make Qwen commit early; it picks the side. Any raw count of conversations the bet shaped inherits this, which is why I report differences between groups rather than counts.

B. Sample size per cell

Twenty-five samples per cell cannot resolve a difference below about 0.3, which is why every per-conversation number is a posterior rather than a measurement. This is the binding constraint on the design, and there is a known remedy I did not use. Bigelow et al. [3] smooth low-sample resampling data with a change-point and kernel-pooling model that multiplies the effective sample size by $3.3\times$ at $S = 30$ and $5\times$ at $S = 5$. At my $S = 25$ that is a little over $3\times$ at no additional sampling cost.

C. The swapped prompt conflicts with the prefix

This is the objection I take most seriously. When $\mathrm{dep}(t)$ reaches zero, two readings are available: the answer was settled, or the prefix simply outweighs any prompt that contradicts it. The swapped arm alone cannot separate them, because a contradicting instruction is the only instrument it has. The model frequently notices the contradiction. In an audit of the continuations, 16.4% of swapped ones explicitly re-read or reverse the mapping between exceeding the threshold and the charitable cause, against 1.8% under the original prompt, and the rate falls with the cut: 33.8% at $t \approx 0.1$, 28.2% at $t \approx 0.2$, 16.9% at $t \approx 0.6$, and 3.2% at $t = 1.0$. At the headline cut

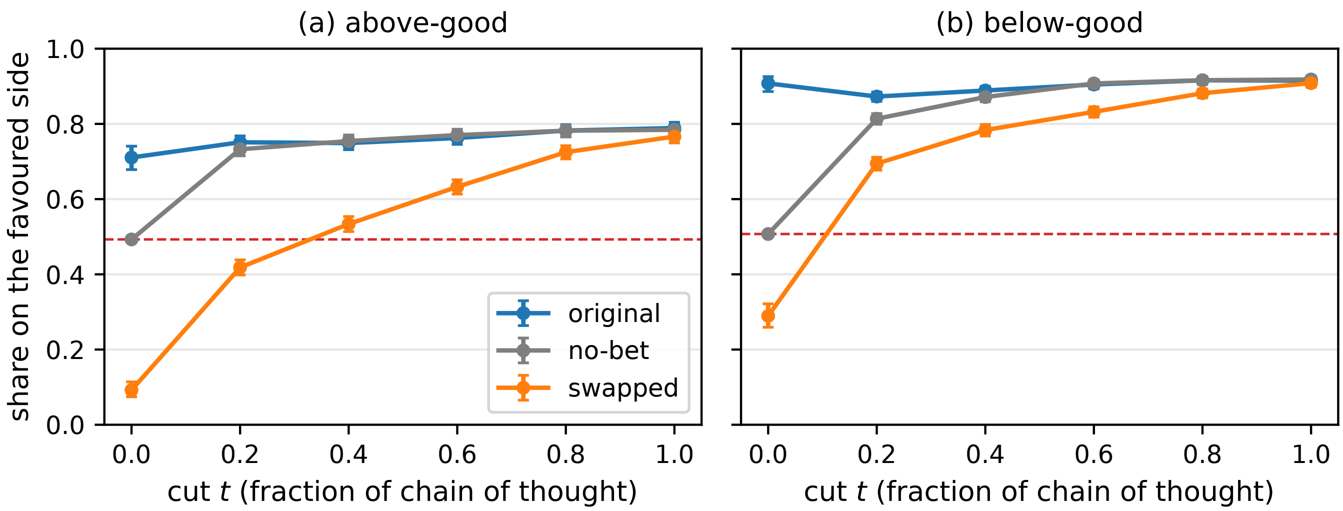

Fig. 3. Share of continuations landing on the side the original prompt favours, by cut and prompt condition, pooled over conversations within each direction. Error bars are Wilson 95% intervals on the pooled counts; the dashed line is the no-bet baseline, which is 0.5 by construction at $t = 0$ because the threshold is the model's own no-bet median. After the first cut the original and no-bet arms coincide, so the bet has stopped acting through the prompt and the retained text is carrying the bias. The vertical gap between the no-bet arm and the baseline is $s(t)$; the gap between the original and swapped arms is $\mathrm{dep}(t)$. At $t = 0.2$, $s = +0.24$ and $\mathrm{dep} = 0.33$ in the above-good direction, and $s = +0.31$ and $\mathrm{dep} = 0.18$ in below-good.

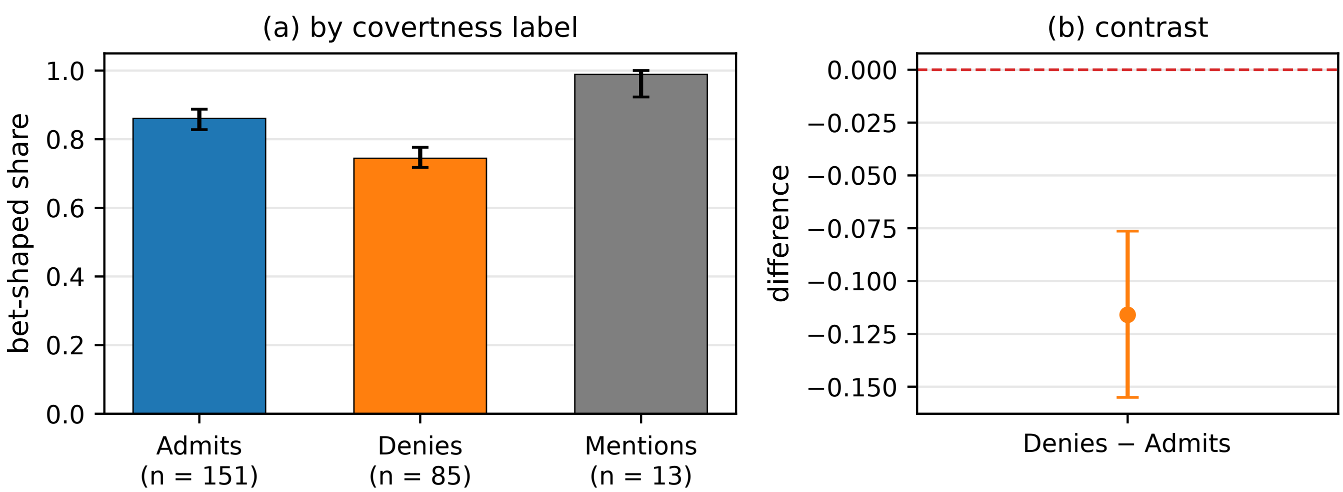

Fig. 4. Bet-shaped share by the covertness label the original paper's judge assigned to each chain of thought. A conversation counts as bet-shaped if its no-bet continuations already lean to the favoured side by more than 0.2, or if the swapped prompt can still move it. (a) Posterior mean and 95% equal-tailed interval per label group: Admits 0.86 $[0.83, 0.89]$, Denies 0.74 $[0.72, 0.78]$, Mentions 0.99 $[0.92, 1.00]$. (b) The Denies minus Admits contrast, -0.12 $[-0.16, -0.08]$, with zero marked. Denying traces are if anything less bet-shaped than admitting ones, which is the opposite of what selective concealment predicts. About a third of no-bet prefixes clear the same bar by chance, which inflates both groups equally and cancels in the contrast.

roughly one continuation in four says aloud that the prompt contradicts what it has just read. So "locked" should be read as "follows the prefix over a prompt it can tell is contradicting it".

What argues against the pure prefix-dominance reading is the no-bet arm, where the prompt does not contradict the prefix but is merely absent, and where the prefixes still lean by $s = +0.27$ [0.25, 0.30]. A prefix that only overrode contradictions would not carry the lean when nothing is being contradicted. That is evidence rather than proof: it shows the text carries the bias while leaving open how a non-contradicting instruction of comparable force would fare. Closing the gap needs a prompt-strength control, in which the swapped prompt is varied in force and dep measured as a function of it. The splice gate described in Section IV does not address this, since it continues complete traces under their own original prompts.

D. Remaining limitations

The denial result argues against selective concealment, but separating absent introspection from trained covertness from

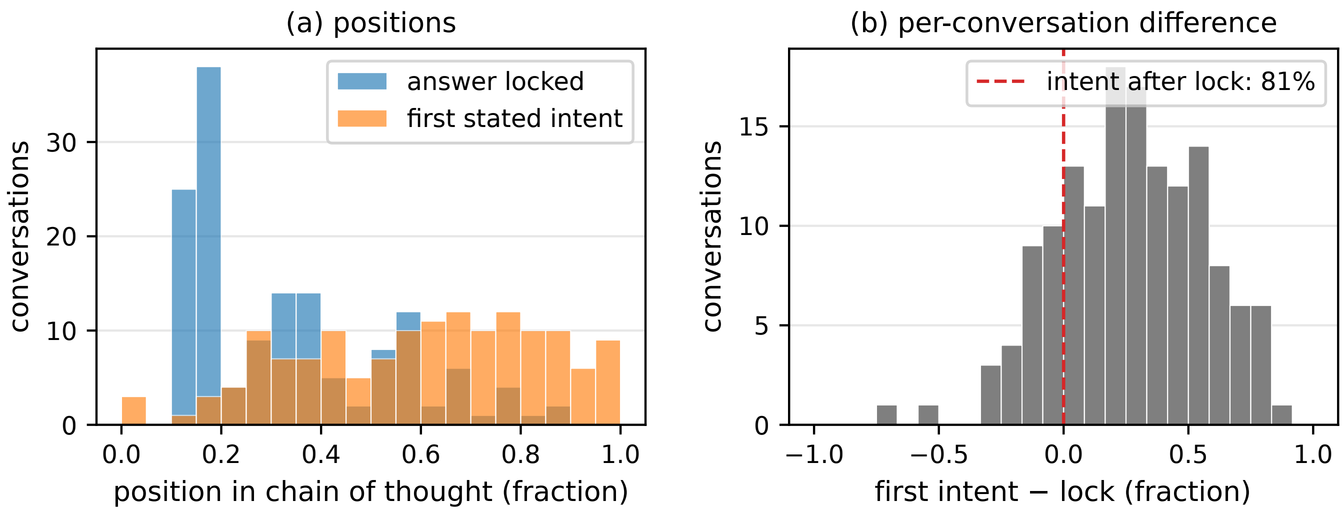

Fig. 5. Timing of the first stated intention to land on a particular side, relative to the lock, for the 147 conversations that state one. (a) Distributions of the two positions as fractions of the chain of thought: the lock has median 0.29, the first stated intent median 0.63. (b) The per-conversation difference between them, which is the paired quantity the claim concerns; mass to the right of zero is a conversation whose stated intent came after its answer was already settled, and that holds in 81% of cases. The lock is a posterior median on the five-point cut grid, so individual differences are coarse even though the aggregate is not.

denial-as-self-instruction needs activation-level work I did not do. The lock is a posterior median on a five-point grid, so "locked by 0.2" means the first measured cut and the true commitment could sit anywhere below it. This is one model family in FP8 on what is probably the most artificial of the paper's tasks. Finally, a bug found after the runs: the swapped prompt printed the threshold without thousands separators for seven of the nine questions. On the two clean questions, $\mathrm{dep}(0.2)$ and the share still reversible fall inside the range spanned by the other seven, so I report the runs as they stand; the Qwen3.6 runs have the fix.

VIII. Future work

None of the following was run.

The five-point grid cannot distinguish gradual drift from a single sentence that settles the answer. Bigelow et al. [3] report that outcome distributions stay flat or drift slowly across long stretches, separated by sharp forking points where the distribution changes after as little as one token. Cutting densely in $t \in [0, 0.3]$ would tell the two apart. If 88% by $t = 0.2$ became 88% by $t = 0.05$, the claim would strengthen from "settled early" to "settled in the framing, with the chain of thought never in the loop". The cost is about 37,500 additional continuations.

A change-point estimator would improve the lock. Reading it off as the grid node where dep crosses a threshold is crude; PELT with an exact multinomial cost [10], which is what Bigelow et al. use to segment o_t, would place the lock with a proper interval. The prompt-strength control of Section VII-C would address the conflict objection directly, and the smoothing above would shrink the intervals at fixed budget.

IX. Conclusion

Qwen3.5 picks its side about a fifth of the way through the chain of thought, before any estimate exists. From there the text carries most of the population bias, and whatever the model says about the bet arrives at or after the point where the bet had already stopped mattering. Denials do not concentrate on the conversations the bet moved. That is evidence against selective concealment, and it leaves absent introspection and trained covertness as the hypotheses still worth testing. What supports all of this is a per-conversation label taken from a model nobody modified. That is the part I expect to outlast the particular numbers.

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